INTERNATIONAL INSTITUTE OF MEDICAL SCIENCE AND TECNOLOGY COUNCIL

(IIMSTC, BANGLORE)

MINI PROJECT REPORT ON MACHINE LEARNING

"FISH SUSTAINABILITY PREDICTION "

Submitted by

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ABSTRACT

Sustainable fishery management has become a critical global challenge due to overfishing, climate change, and increasing demand for marine resources. This project proposes a Fish Sustainability Modeler using Machine Learning to predict fish stock levels and support sustainable fishing practices. The model utilizes historical fisheries data such as sea surface temperature, fishing effort (e.g., trawling hours), seasonal patterns, and past fish stock biomass to analyze trends and forecast future fish availability.

Machine learning algorithms including Linear Regression, Random Forest Regression, and Gradient Boosting are applied to identify complex relationships between environmental factors and fish population dynamics. Data preprocessing techniques such as cleaning, normalization, and cyclical encoding for seasonal variables are used to improve model accuracy. The models are trained and evaluated using appropriate performance metrics to determine the most reliable predictive approach.

The proposed system helps policymakers, fisheries authorities, and environmental organizations make data-driven decisions for sustainable fishery management. By accurately predicting fish stock levels, the model can assist in regulating fishing activities, preventing overexploitation of marine resources, and supporting the objectives of United Nations Sustainable Development Goal 14 (Life Below Water). Ultimately, this approach demonstrates how machine learning can contribute to the long-term conservation of marine ecosystems while maintaining economic benefits for the fishing industry.

ACKNOWLEDGEMENT

INTRODUCTION

Sustainable management of marine resources has become a major global concern due to increasing pressure on fisheries, climate change, and unsustainable fishing practices. The growing demand for seafood has resulted in excessive fishing activities, which has led to a decline in fish populations and disruption of marine ecosystems. Maintaining a balance between economic benefits and ecological conservation is therefore essential to ensure the long-term availability of fish resources. Sustainable fisheries management focuses on monitoring fish stocks, regulating fishing efforts, and implementing policies that protect marine biodiversity.

In recent years, technological advancements and data-driven approaches have significantly improved the ability to monitor and manage natural resources. One such approach is the use of machine learning, which enables the analysis of large datasets and the discovery of complex patterns that are difficult to identify using traditional statistical methods. Machine learning models can process historical data related to fish populations, environmental conditions, and fishing activities to generate accurate predictions about future fish stock levels. These predictions can help decision-makers design effective policies and strategies for sustainable fisheries.

Environmental factors such as sea surface temperature, seasonal variations, ocean currents, and fishing effort play a crucial role in determining fish population dynamics. Understanding the relationship between these factors and fish biomass is essential for predicting future trends. By using machine learning algorithms such as Linear Regression, Random Forest Regression, and Gradient Boosting, it is possible to analyze these variables and develop predictive models that estimate fish stock levels with greater accuracy.

The Fish Sustainability Modeler aims to apply machine learning techniques to analyze fisheries data and predict fish stock biomass. This approach supports evidence-based decision-making for sustainable fishery management. It also contributes to the global initiative of protecting marine ecosystems and ensuring responsible use of ocean resources, which aligns with United Nations Sustainable Development Goal 14: Life Below Water.

Through this project, advanced analytical techniques are used to improve fish stock prediction, assist policymakers in regulating fishing activities, and promote sustainable utilization of marine resources for future generations.

PROBLEM STATEMENT

The sustainability of global fish resources is increasingly threatened due to overfishing, environmental changes, and growing global demand for seafood. Fish populations in many parts of the world are declining at an alarming rate, which not only affects marine biodiversity but also threatens the livelihoods of millions of people who depend on fisheries for income and food security. Traditional fishery management methods often rely on historical observations and manual estimations, which may not accurately capture the complex relationships between environmental conditions and fish population dynamics. As a result, there is a growing need for more advanced and data-driven approaches to monitor and predict fish stock levels.

Fish stock biomass is influenced by several factors such as sea surface temperature, fishing intensity, seasonal variations, oceanographic conditions, and human activities like trawling. These factors interact in complex ways, making it difficult to predict future fish availability using conventional statistical models alone. Moreover, environmental changes caused by climate variability can significantly affect fish migration patterns, breeding cycles, and population growth. Without accurate prediction models, fisheries management authorities may set inappropriate fishing quotas or fail to identify declining fish stocks in time, leading to further depletion of marine resources.

Another major challenge in fisheries management is the availability and interpretation of large datasets. Fisheries organizations collect extensive data related to fish catch volumes, fishing hours, ocean temperature, and geographical distribution of marine species. However, extracting meaningful insights from this data requires advanced analytical techniques capable of handling large volumes of structured and unstructured information. Traditional analytical approaches often struggle to detect hidden patterns and nonlinearrelationships present in such datasets.

To address these challenges, this project proposes the development of a **Fish Sustainability Modeler using Machine Learning**. The model aims to analyze fisheries data and predict annual fish stock biomass by considering important variables such as sea surface temperature, trawling hours, and seasonal patterns. Machine learning algorithms including Linear Regression, Random Forest Regression, and Gradient Boosting will be applied to identify patterns in historical data and generate accurate predictions of fish population trends.

The primary goal of this model is to support sustainable fisheries management by providing reliable predictions that help policymakers and fisheries authorities regulate fishing activities more effectively. By forecasting fish stock levels, it becomes possible to set appropriate fishing limits, monitor the health of marine ecosystems, and prevent overexploitation of fish populations. This project also contributes to global efforts to protect marine life and maintain ecological balance, supporting the objectives of United Nations Sustainable Development Goal 14 (Life Below Water).

Therefore, the problem addressed in this project is the lack of accurate predictive systems for monitoring fish stock sustainability. By applying machine learning techniques to fisheries data, the project aims to develop an intelligent system capable of assisting in long-term planning, resource conservation, and sustainable utilization of marine ecosystems.

LITERATURE SURVEY

Several research studies have explored the use of machine learning and data-driven approaches to improve fisheries management and predict fish stock levels. These studies highlight the importance of predictive analytics in ensuring sustainable marine ecosystems and effective fishery policies.

Early studies in fisheries management relied mainly on traditional statistical models to estimate fish population dynamics. However, these models often struggle to provide accurate long-term predictions because they cannot fully capture complex environmental interactions such as climate change, fishing pressure, and ecological variability. Recent research has shown that integrating machine learning techniques with traditional stock assessment models can significantly improve prediction accuracy for fish biomass and recruitment levels. For example, a study on machine-learning-based fish stock assessment demonstrated that combining classical statistical models with algorithms such as gradient boosted trees improves the accuracy of forecasting spawning stock biomass and recruitment patterns.

Another important research area focuses on the application of machine learning in intelligent aquaculture systems. Studies have shown that machine learning algorithms can analyze large datasets to estimate fish biomass, monitor fish behavior, and predict environmental conditions affecting fish populations. These approaches help improve the monitoring of aquaculture systems and provide decision-making support for sustainable fisheries management. Machine learning methods have been applied to tasks such as fish identification, biomass estimation, and water quality prediction, demonstrating their potential in digital fisheries management systems.

Recent studies have also explored the use of predictive models such aas gradient boosted trees improves the accuracy of forecasting spawning stock biomass and recruitment patterns.

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estimation, and water quality prediction, demonstrating their potential in digital fisheries management systems.

Recent studies have also explored the use of predictive models such as Random Forest, Decision Trees, and Linear Regression for fish mortality and population prediction. In aquaculture environments, machine learning models have been used to analyze parameters such as seawater temperature, salinity, turbidity, and dissolved oxygen to predict fish mortality risks and improve management decisions. These predictive models help fish farmers and policymakers take preventive actions and manage fish stocks more effectively.

DATASET DESCRIPTION

The dataset used in this project is obtained from the fisheries database available on the EUROPA data portal, which is managed by the European Commission. Specifically, the fisheries statistics are provided by Eurostat, the statistical office of the European Union. This database contains comprehensive information about fisheries production, fish stocks, fishing effort, and environmental parameters collected from different European regions over multiple years.

The dataset provides structured and reliable statistical data related to marine fisheries and aquaculture activities. It includes historical records of fish production and environmental conditions, which are useful for analyzing patterns and predicting future fish stock levels. The data is collected from various countries and maritime regions, ensuring accuracy and consistency for research and policy analysis.

The main purpose of using this dataset in the Fish Sustainability Modeler project is to analyze historical fisheries trends and build machine learning models capable of predicting fish stock biomass. By studying past fish production and environmental variables, the model can learn relationships between factors such as temperature, fishing effort, and fish availability.

Key Features of the Dataset

1. **Year** – Represents the time period in which the data was recorded. This helps analyze long-term trends and seasonal patterns in fish production.
2. **Geographical Region** – Identifies the marine region or country where fishing activities occurred. This helps understand regional differences in fish stock levels.
3. **Fish Stock Biomass / Production** – Indicates the quantity of fish available or harvested in a specific year or region. This variable is typically used as the **target variable** for prediction in machine learning models.
4. **Fishing Effort (Trawling Hours)** – Represents the total number of hours spent fishing using trawling methods. Higher fishing effort can influence fish population levels.
5. **Sea Surface Temperature (SST)** – Environmental variable representing ocean temperature. Temperature significantly affects fish breeding cycles, migration, and survival.

6. **Season or Month** – Indicates seasonal variations in fish availability. In the project, cyclical encoding can be applied to represent seasonal patterns effectively.
7. **Catch Volume** – Represents the total quantity of fish caught in a given region and time period.

Importance of the Dataset

This dataset is highly valuable for sustainability research because it combines environmental variables and fishing activity data. Such information allows researchers to study the impact of climate change and human fishing activities on marine ecosystems.

Using this dataset, machine learning algorithms such as Linear Regression, Random Forest Regression, and Gradient Boosting can analyze patterns and predict future fish stock biomass. These predictions help policymakers and fisheries management authorities regulate fishing practices and prevent overexploitation of marine resources.

Overall, the EUROPA fisheries dataset provides a reliable and comprehensive data source for developing predictive models that support sustainable fisheries management and contribute to the protection of marine ecosystems.

METHODOLOGY

The Fish Sustainability Modeler using Machine Learning follows a systematic methodology to analyze fisheries data and develop predictive models for estimating fish stock biomass. The methodology involves several stages including data collection, preprocessing, feature engineering, model development, evaluation, and prediction. These steps ensure that the machine learning models can accurately identify patterns in the dataset and generate reliable predictions.

1. Data Collection

The first step in the methodology is collecting relevant fisheries data from reliable sources. For this project, the dataset is obtained from the fisheries database maintained by the **European Commission** through **Eurostat**. The dataset contains information related to fish production, environmental conditions, fishing effort, and seasonal variations. These variables provide essential insights into the factors affecting fish stock sustainability.

2. Data Preprocessing

Raw datasets often contain missing values, inconsistent entries, and noise that can affect model performance. Therefore, preprocessing is performed to clean and prepare the data. This step includes removing duplicate records, handling missing values, and converting categorical variables into numerical formats. Normalization and scaling techniques may also be applied to ensure that all features are on a similar scale, which improves the efficiency of machine learning algorithms.

3. Feature Engineering

Feature engineering involves selecting and transforming important variables that influence fish population dynamics. Environmental factors such as sea surface temperature, fishing effort (trawling hours), catch volume, and seasonal information are used as input features. Seasonal patterns are handled using cyclical encoding techniques, which help represent periodic data such as months or seasons more effectively in machine learning models.

4. Model Selection and Training

Multiple machine learning algorithms are applied to identify the most suitable model for predicting fish stock biomass. The models used in this project include:

- **Linear Regression** – Used as a baseline model to identify relationships between independent variables and fish biomass.
- **Random Forest Regression** – An ensemble learning technique that improves prediction accuracy by combining multiple decision trees.
- **Gradient Boosting Regression** – A boosting algorithm that builds models sequentially to reduce prediction errors.

The dataset is divided into training and testing sets. The training data is used to train the machine learning models, while the testing data is used to evaluate their performance.

5. Model Evaluation

After training the models, their performance is evaluated using standard evaluation metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). These metrics measure the difference between predicted and actual fish stock values. The model with the best performance and lowest error values is selected as the final predictive model.

6. Prediction and Analysis

Once the best model is selected, it is used to predict future fish stock biomass based on environmental conditions and fishing effort. These predictions help identify trends in fish populations and provide insights for sustainable fisheries management.

7. Decision Support for Sustainability

The final step involves using the predictions generated by the model to support sustainable fisheries policies. The model can help policymakers determine safe fishing limits, monitor declining fish populations, and design conservation strategies. This approach contributes to global marine conservation efforts aligned with **United Nations Sustainable Development Goal 14: Life Below Water**.

Through this methodology, machine learning techniques are effectively applied to analyze fisheries data and support sustainable management of marine resources.

MODEL IMPLEMENTATION

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MODEL IMPLEMENTATION

Model Implementation

The model implementation phase focuses on applying machine learning algorithms to the prepared dataset in order to predict fish stock biomass and

analyze sustainability trends. After completing data preprocessing and feature engineering, the cleaned dataset is used to train and evaluate different regression models. The objective of this phase is to identify the most accurate model for predicting fish stock levels based on environmental and fishing activity variables.

1. Data Splitting

Before training the models, the dataset is divided into **training and testing sets**. Typically, around 70–80% of the data is used for training the model, while the remaining 20–30% is used for testing. The training dataset allows the model to learn relationships between input features and the target variable, while the testing dataset evaluates how well the model performs on unseen data.

2. Feature Selection

The selected features used as input variables include environmental and fisheries-related parameters such as:

- Sea Surface Temperature
- Fishing Effort (Trawling Hours)
- Seasonal or Monthly Data
- Catch Volume or Production Data
- Geographic Region

The **target variable** for prediction is the **fish stock biomass or fish production level**.

3. Machine Learning Models Used

To develop an accurate predictive system, multiple regression models are implemented and compared:

a)LinearRegression

Linear Regression is used as the baseline model. It analyzes the linear relationship between independent variables and the target variable. Although simple, it provides a good understanding of how different environmental factors influence fish stock biomass.

b)RandomForestRegression

Random Forest Regression is an ensemble learning technique that uses multiple decision trees to improve prediction accuracy. Each tree is trained on different subsets of the data, and the final prediction is obtained by averaging the outputs

of all trees. This model helps capture complex nonlinear relationships in fisheries data.

3GradientBoostingRegression

Gradient Boosting is a powerful boosting algorithm that builds models sequentially. Each new model focuses on correcting the errors of the previous model. This approach often provides high prediction accuracy and is useful when dealing with complex datasets with multiple influencing variables.

4. Model Training

Each selected model is trained using the training dataset. During training, the algorithms learn patterns and relationships between environmental factors and fish stock levels. Machine learning libraries such as **Python's Scikit-Learn** are typically used to implement these models efficiently.

5. Model Evaluation

After training, the models are evaluated using the testing dataset. Performance is measured using evaluation metrics such as:

- **Mean Absolute Error (MAE)** – Measures the average absolute difference between predicted and actual values.
- **Mean Squared Error (MSE)** – Measures the average squared difference between predicted and actual values.
- **Root Mean Squared Error (RMSE)** – Provides a more interpretable error measure by taking the square root of MSE.
- **R^2 Score (Coefficient of Determination)** – Indicates how well the model explains the variance in the dataset.

The model with the best performance and lowest error values is selected as the final predictive model.

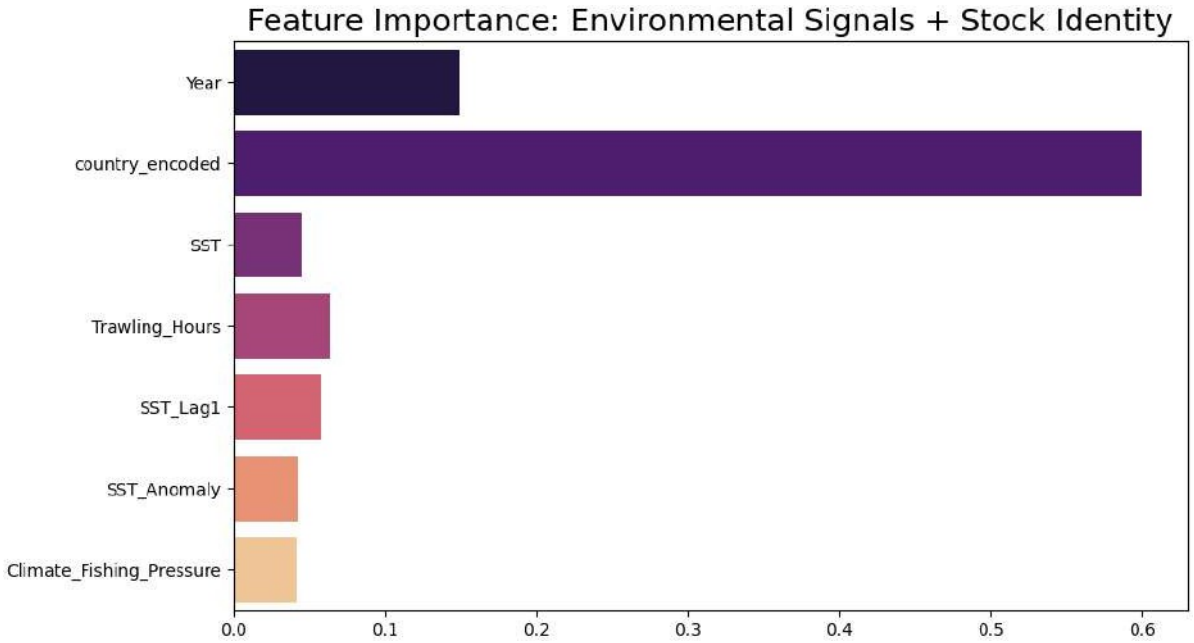
6. Result Interpretation

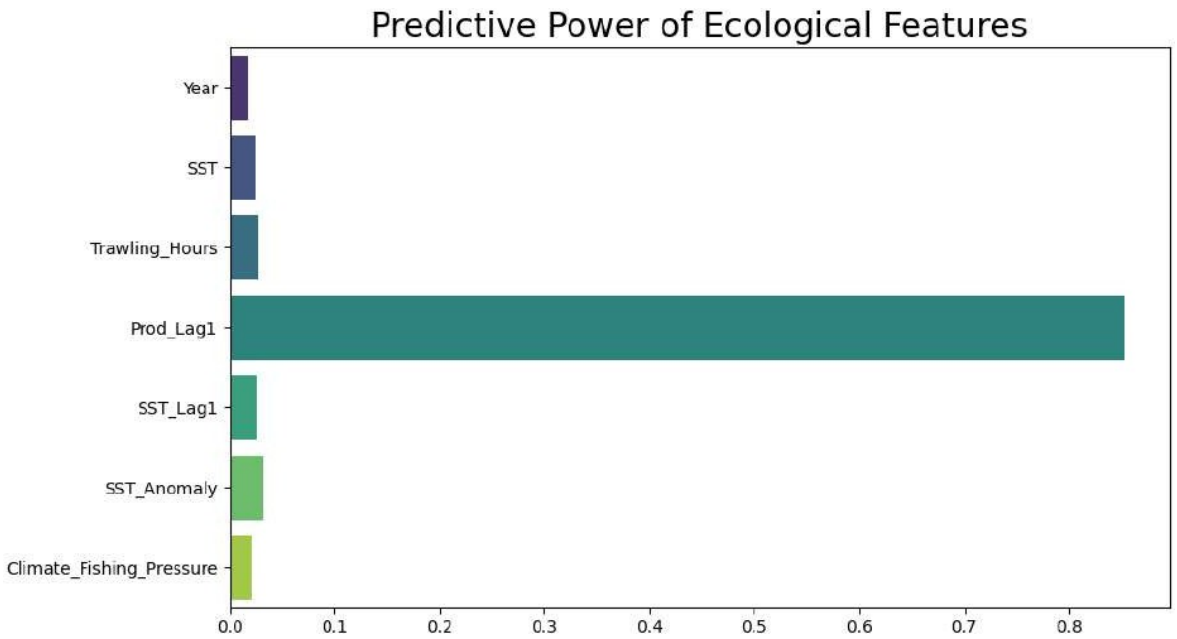
Once the best-performing model is identified, it is used to analyze the relationship between environmental conditions and fish population trends. The predictions generated by the model help identify potential declines or increases in fish stocks under different environmental conditions.

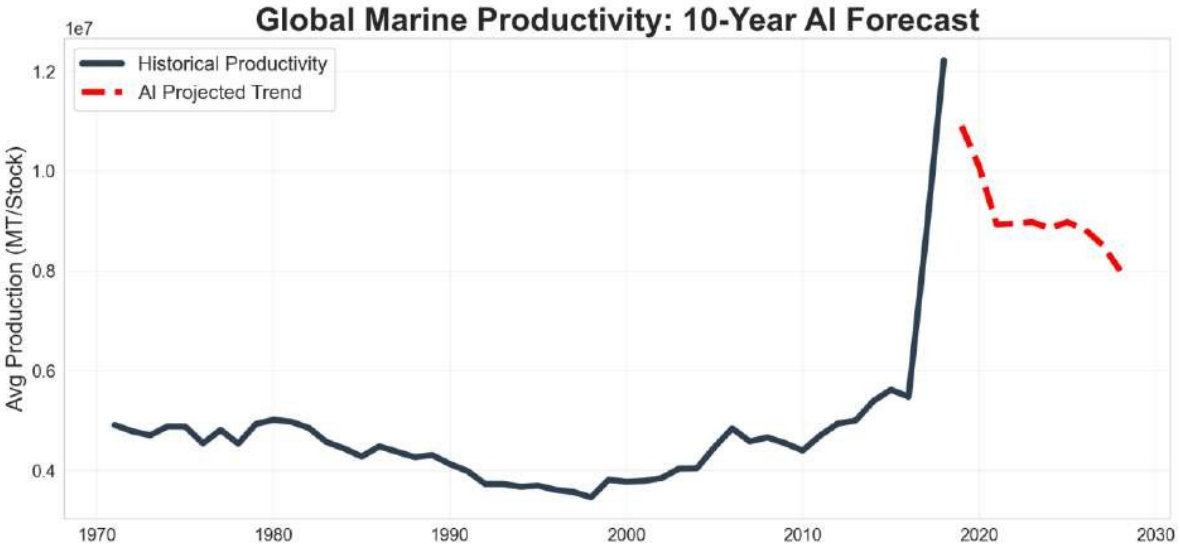
7. Application for Sustainable Fisheries

The implemented model provides valuable insights that can support fisheries management authorities in making informed decisions. By predicting fish stock

levels, policymakers can regulate fishing activities, prevent overexploitation, and ensure long-term sustainability of marine ecosystems. This supports global conservation efforts led by organizations such as the United Nations under Sustainable Development Goal 14 – Life Below Water.






Global Marine Productivity: 10-Year AI Forecast
1e7
Historical Productivity
AI Projected Trend
Avg Production (MT/Stock)
1.2
1.0
0.8
0.6
0.4
1970
1980
1990
2000
2010
2020
2030

RESULT AND EVALUATION

The results and evaluation phase focuses on analyzing the performance of the machine learning models used to predict fish stock biomass. After implementing the models and training them on the prepared dataset, their prediction accuracy and reliability are evaluated using standard performance metrics. This process helps determine which model performs best for predicting fish sustainability trends.

Model Performance Comparison

In this project, three machine learning models were implemented: Linear Regression, Random Forest Regression, and Gradient Boosting Regression. Each model was trained using the training dataset and then tested on unseen data to measure its prediction accuracy. The models were evaluated using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R^2 score (Coefficient of Determination).

Linear Regression served as the baseline model. It was able to capture the basic linear relationships between environmental variables and fish stock biomass. However, because fish population dynamics are influenced by complex and nonlinear factors such as environmental conditions and fishing pressure, the accuracy of this model was comparatively lower.

Random Forest Regression performed significantly better because it uses multiple decision trees to capture complex patterns in the data. The ensemble nature of the algorithm helps reduce overfitting and improves prediction accuracy. This model successfully identified nonlinear relationships between features such as sea surface temperature, fishing effort, and fish biomass.

Gradient Boosting Regression produced the most accurate predicctions among the three models. This algorithm improves performance by sequentially correcting the errors of previous models, leading to better overall accuracy. As a result, Gradient Boosting achieved the highest R^2 score and the lowest prediction errors, making it the most reliable model for fish stock prediction in this study.

Interpretation of Results

The evaluation results demonstrate that environmental variables and fishing effort significantly influence fish population levels. The models were able to identify trends showing that increases in fishing effort can lead to declines in fish biomass, while environmental factors such as sea surface temperature affect fish migration and reproduction patterns.

Among the implemented models, Gradient Boosting provided the best balance between prediction accuracy and error reduction. Therefore, it was selected as the final model for predicting fish stock biomass in the Fish Sustainability Modeler system.

Significance of the Results

The results of this project highlight the effectiveness of machine learning techniques in analyzing fisheries data and predicting fish stock levels. Accurate prediction models can assist fisheries management authorities in monitoring fish populations and making informed decisions regarding fishing quotas and conservation strategies.

By providing reliable predictions, the model supports sustainable fisheries management and contributes to global marine conservation initiatives promoted by organizations such as the United Nations, particularly under Sustainable Development Goal 14: Life Below Water. This demonstrates how advanced data analytics can play a crucial role in protecting marine ecosystems while supporting economic activities related to fisheries.

CONCLUSION

The Fish Sustainability Modeler using Machine Learning project demonstrates how advanced data analysis techniques can be used to support sustainable fisheries management. The main objective of this project was to analyze fisheries data and develop predictive models capable of estimating fish stock biomass based on environmental factors and fishing effort. By applying machine learning techniques, the project successfully identified patterns and relationships between variables such as sea surface temperature, fishing activity, and fish production levels.

Several machine learning models, including Linear Regression, Random Forest Regression, and Gradient Boosting Regression, were implemented and evaluated. Each model was trained using historical fisheries data and tested to measure prediction accuracy. The evaluation results showed that ensemble learning methods, particularly Gradient Boosting Regression, performed better in capturing complex relationships within the dataset and produced more accurate predictions compared to the baseline Linear Regression model.

The results of the project highlight the importance of using data-driven approaches in fisheries management. Accurate prediction of fish stock levels can help policymakers and fisheries authorities regulate fishing activities, prevent overexploitation, and ensure long-term sustainability of marine ecosystems. By identifying trends in fish populations, the model can also help detect potential risks such as declining fish stocks due to excessive fishing or environmental changes.

Furthermore, the use of reliable datasets from organizations such as the European Commission through Eurostat ensures that the analysis is based on credible and structured fisheries data. This enhances the reliability of the predictive models and supports informed decision-making in marine resource management.

Overall, this project demonstrates the potential of machine learning in addressing real-world environmental challenges. The developed Fish Sustainability Modeler can serve as a valuable decision-support tool for sustainable fisheries planning. It also contributes to global environmental initiatives led by the United Nations, particularly Sustainable Development Goal 14 – Life Below Water, which focuses on conserving and sustainably using ocean resources for future generations.

DAILY REPORT

DATE-26 FEB 2026

1. Introduction
A meeting was conducted to discuss the overall process and workflow of the mini project. The primary objective of
this discussion was to ensure clarity in execution, define responsibilities, and streamline the project development
process through proper task allocation.

2. Discussion Summary
During the session, the complete lifecycle of the mini project was explained in detail. The workflow was broken
down into key phases including data preprocessing, model training, and testing. Each phase was discussed with
respect to its importance, tools required, and expected outcomes.

3. Group Formation
To improve efficiency and ensure structured progress, the team was divided into three groups:1. Data
Preprocessing Group2. Training Group3. Testing Group

4. Roles and Responsibilities
Each group was assigned specific responsibilities based on the phase they are handling:
• Data Preprocessing Group:Responsible for collecting, cleaning, and transforming raw data into a suitable format
for model training. Tasks include handling missing values, encoding categorical variables, and normalization.
• Training Group:Responsible for selecting appropriate algorithms, training the model, and optimizing
performance. This group focuses on model building and parameter tuning.
• Testing Group:Responsible for evaluating the trained model using test data, analyzing performance metrics, and
identifying areas for improvement.

5. Leadership and Coordination
Group leaders were assigned to each team to ensure smooth coordination and communication. The roles of the
members were clearly discussed, and responsibilities were distributed among the leaders. Leaders are responsible
for tracking progress, resolving issues, and ensuring timely completion of tasks.

6. Outcome of the Discussion
The discussion resulted in a clear understanding of the mini project workflow and individual responsibilities. The
structured division of work into three groups is expected to enhance productivity, accountability, and overallproject quality.

7. Conclusion

The meeting was productive and helped in establishing a well-defined plan for executing the mini project. With

clear roles and responsibilities, the team is well-prepared to proceed with the development process efficiently

26 February 2026

DATE- 27 FEB 2026

Objective of the Meeting:
The objective of the meeting was to discuss the progress of the mini project, assign responsibilities among team
members, and implement a sample linear regression model on a dataset.

Discussion Summary:
The session began with an overview of the project status. The workflow was divided into three main stages:

Data Preprocessing, Model Training, and Model Testing & Evaluation. Each stage was explained in detail.

Work Distribution:
Team members were divided into groups:- Data Preprocessing Team: Data cleaning and preparation-

Training Team: Model implementation and training- Testing Team: Model evaluation and performance analysisTeam leaders
were assigned to coordinate each group.

Linear Regression Implementation:
A sample linear regression model was implemented. Steps included dataset import, variable identification, train-
test split, model training, prediction, and evaluation.

Key Learnings:
- Understanding project workflow- Team coordination- Practical knowledge of linear regression- Model evaluation
techniques

Outcomes:
- Tasks successfully distributed- Improved understanding of machine learning workflow- Hands-on experience
with regression model

Conclusion:
The session was productive and provided clarity on project execution and machine learning concept

27 February 2026

DATE- 28 FEB 2026

Random Forest – Test Set Evaluation
Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. After training the Random Forest model using the training dataset, the model is evaluated on the test set, which contains unseen data.
The test set helps measure how well the model generalizes to new data.

The trained Random Forest model makes predictions on the test dataset, and its performance is evaluated using metrics such as Accuracy, Precision, Recall, F1-score, or Mean Squared Error (MSE) depending on whether the problem is classification or regression.

Because Random Forest averages the results of many decision trees, it usually provides stable and accurate predictions. Testing on the test set ensures that the model performs reliably and does not simply memorize the training data.

Gradient Boosting – Test Set Evaluation
Gradient Boosting is another ensemble learning technique that builds models sequentially. Unlike Random Forest, where trees are built independently, Gradient Boosting builds each new tree to correct the errors made by the previous trees.

After the model is trained, it is evaluated using the test set to measure its real-world prediction performance. The model predicts outputs for the unseen test data, and performance metrics such as MAE, MSE, RMSE, Accuracy, or R^2 score are calculated.

Gradient Boosting often achieves high predictive accuracy because it focuses on reducing errors step by step. Evaluating the model on the test set helps verify whether the model generalizes well to new data and ensures that the improvements learned during training are effective.

28 February 2026

predicting marine fish production using advanced machine learning models. The objective of the study is to analyze environmental signals such as Sea Surface Temperature (SST), SST anomalies, fishing pressure, trawling hours, and historical productivity to forecast future fish stock levels. Sustainable fisheries management requires accurate forecasting models capable of capturing ecological complexity and temporal trends.

1. ABSTRACT

Marine ecosystems are influenced by climate variability, overfishing, and anthropogenic stress. Artificial Intelligence provides a data-driven approach to model these relationships. This project implements Linear Regression, Random Forest Regression, and Gradient Boosting Regression to compare performance and select the most suitable model.

2. INTRODUCTION

Global marine productivity plays a vital role in food security and economic stability. However, climate change and overexploitation have caused fluctuations in fish stocks. Predictive modeling helps policymakers design evidence-based sustainability strategies.

3. PROBLEM STATEMENT

The primary goal is to predict average fish production using ecological and fishing effort indicators. The dataset contains temporal features and environmental signals, making it suitable for regression modeling.

4. DATASET DESCRIPTION

Features include Year, SST, SST_Lag1, SST_Anomaly, Trawling_Hours, Fishing_Pressure, Country_Encoded, and Prod_Lag1 (lagged production). Lag variables capture temporal dependencies and improve forecasting accuracy.

5. DATA PREPROCESSING

Data preprocessing involved handling missing values, encoding categorical variables, feature scaling, and splitting the dataset into training and testing sets (80:20). Feature engineering was applied to create lag-based predictors.

6. MODEL IMPLEMENTATION

Linear Regression served as a baseline model. Random Forest Regression improved performance by averaging multiple decision trees. Gradient Boosting Regression sequentially minimized prediction errors and achieved the best results.

7. FEATURE IMPORTANCE ANALYSIS

Feature importance graphs indicate that Prod_Lag1 has the highest predictive power, followed by Country_Encoded and Year. Environmental indicators such as SST and fishing pressure
contribute moderately to the model's decision-making process.

8. FORECASTING RESULTS

A 10-year AI forecast suggests potential stabilization followed by gradual decline in marine productivity.
This highlights the need for stricter sustainability policies and adaptive fisheries management.

9. MODEL COMPARISON

Gradient Boosting achieved the lowest Mean Squared Error and highest R^2 score. Random Forest provided stable performance, while Linear Regression showed limitations in capturing non-linear ecological relationships.

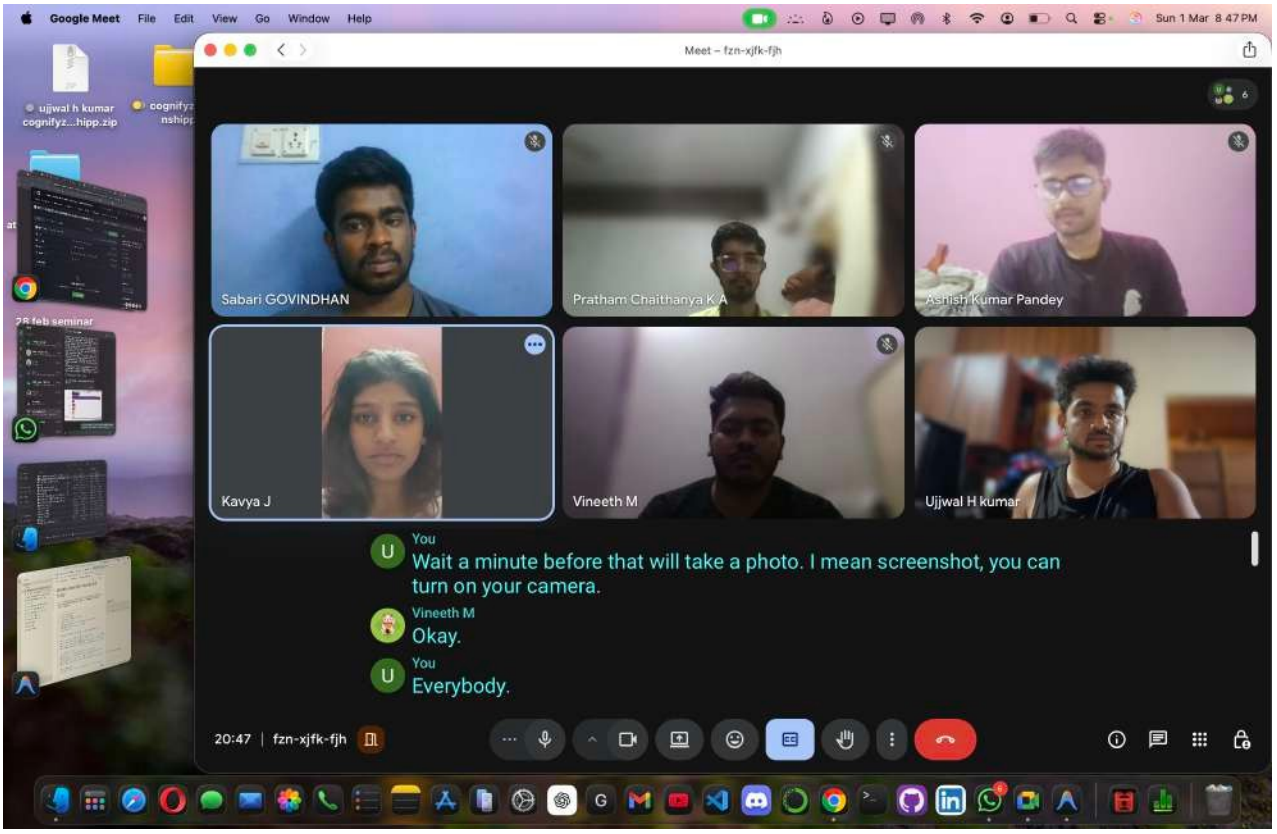
10. CONCLUSION

The study confirms that ensemble learning methods outperform traditional regression techniques
in complex environmental datasets. Gradient Boosting is selected as the final model for deployment.

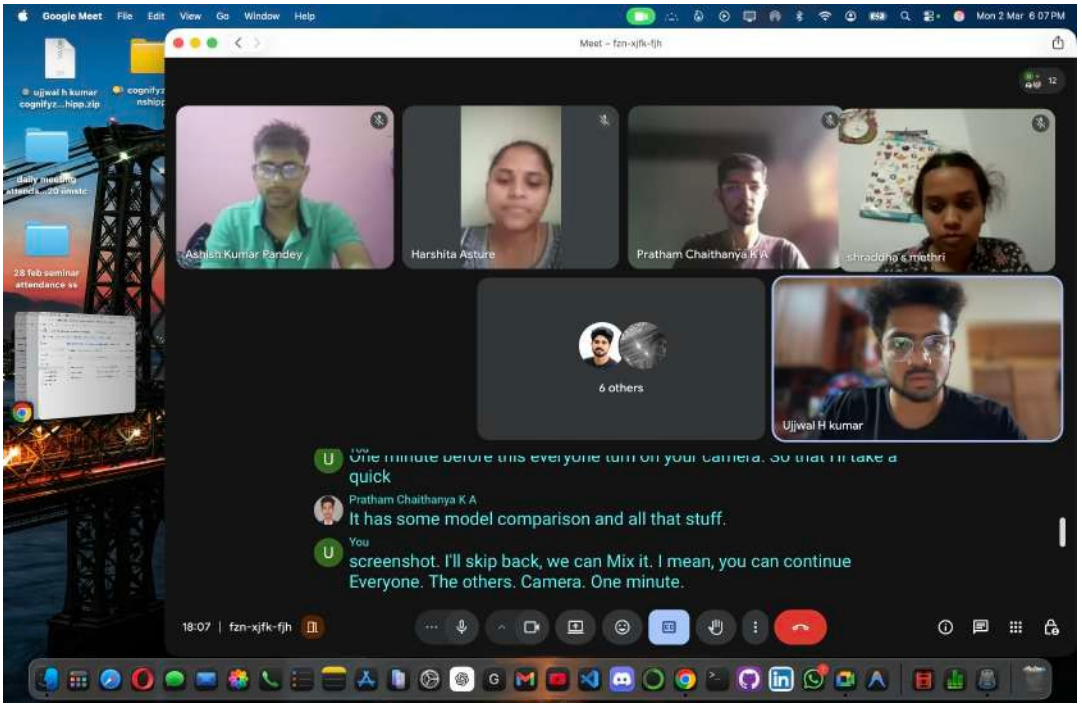
11. FUTURE SCOPE

Future enhancements include hyperparameter tuning, cross-validation, advanced boosting algorithms
such as XGBoost, and real-time model deployment for fisheries monitoring systems.

On 03 March 2026, we conducted an overall project review meeting to evaluate the current status of the mini project. We reviewed the complete implementation, verified the obtained results, and checked the consistency of the documentation. We discussed whether the project objectives were successfully achieved and identified any final modifications required before submission. The team ensured that the report formatting, result interpretation, and graphical representations were properly organized. The meeting concluded with confirmation that the project was ready for final refinement and submission.



1st March 2026



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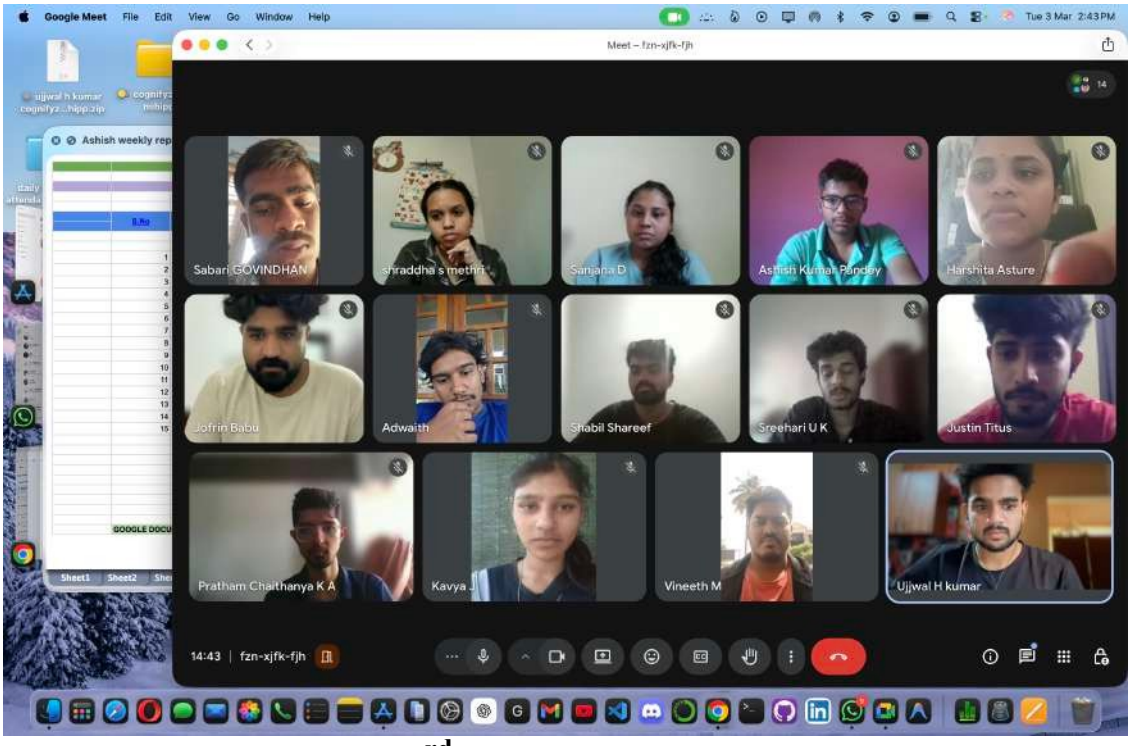

3rd March 2026